

AI-Native Price Mapping Engine for Unstructured Procurement Data

1. Objective

The goal of this solution is to design a layout-agnostic extraction system capable of processing highly unstructured documents and identifying vendor payments, including indirect (“proxy vendor”) relationships such as Action Forms billing scenarios.

2. Problem Context

Traditional rule-based (regex) extraction fails in real-world environments where document formats vary significantly.

Additionally, payments are often routed through intermediaries, making it difficult to identify the actual product or service (e.g., Action Forms).

3. Solution Overview

The solution is designed as a hybrid system combining:

- Structured data pipeline (Medallion Architecture)
- AI-native extraction layer for semantic understanding

4. Architecture

PDF → OCR → TEXT → LLM Layer → Structured Output → Analytics

Explain:

- Bronze → Raw OCR
- Silver → Clean text
- Gold → Price mapping
- LLM Layer → Semantic extraction

5. LLM-Based Extraction (KEY SECTION)

The extraction logic was upgraded from regex-based parsing to an AI-native approach.

Instead of relying on fixed patterns, the system interprets context to extract:

- Vendor of record
- Payment amount
- Service/product
- Relationship to Action Forms

- Proxy vendor relationships

6. Proxy Vendor Detection (VERY IMPORTANT)

The system identifies indirect payment relationships where the vendor of record differs from the actual product provider.

Example:

- Vendor: Global Tech Solutions
- Product: E-Form System
- Inference: Equivalent to Action Forms

This enables detection of “money through the noise” in unstructured datasets.

7. Implementation Approach

The LLM layer is implemented as a model-agnostic component and can integrate with:

- OpenAI GPT
- Google Gemini
- Local LLMs (Ollama)

Due to API constraints during development, the current implementation demonstrates simulated LLM outputs reflecting expected behavior.

8. Sample Output (IMPORTANT)

Show table:

Vendor	Amount	Product	Action Forms	Explanation
ABC Consulting	\$25,000	Action Forms	Yes	Direct implementation
Global Tech Solutions	\$60,000	E-Form System	Yes	Proxy vendor

9. Key Strengths

- Layout-agnostic extraction
- AI-driven reasoning (not rule-based)
- Proxy vendor detection
- Scalable and model-agnostic design